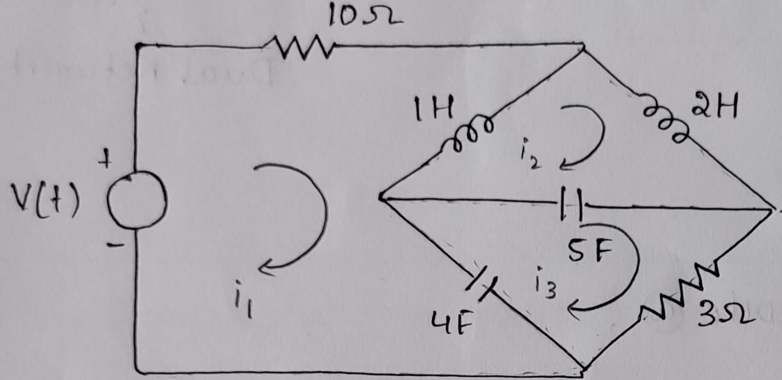
Module-5 unit 3

V. Praharshika [4D]

1. For the network shown in figure, Draw its dual. Write the integro differential form (i) Mesh Equations for the given network (ii) Node equations for the dual $V(t) = 10 \sin 40t$.



(i) KVL for mesh 1,

$$10 = 10i_1 + \frac{1d}{dt}(i_1 - i_2) + \frac{1}{4}\int (i_1 - i_3)\,dt$$

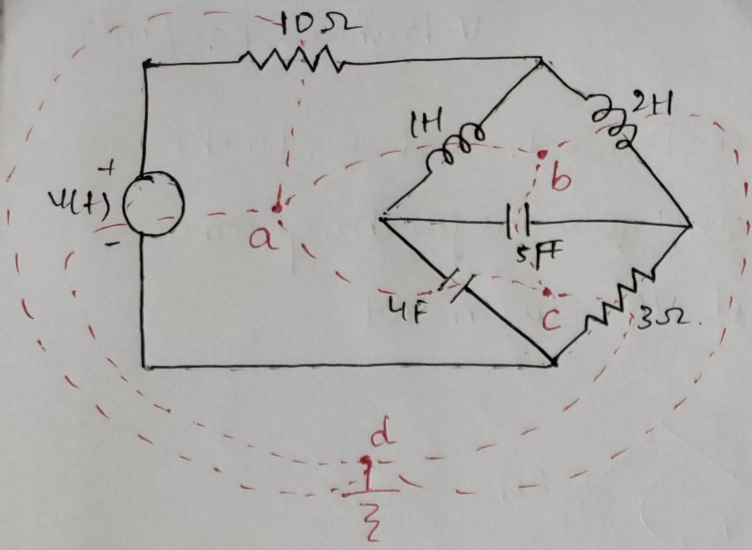
KVL for mesh 2,

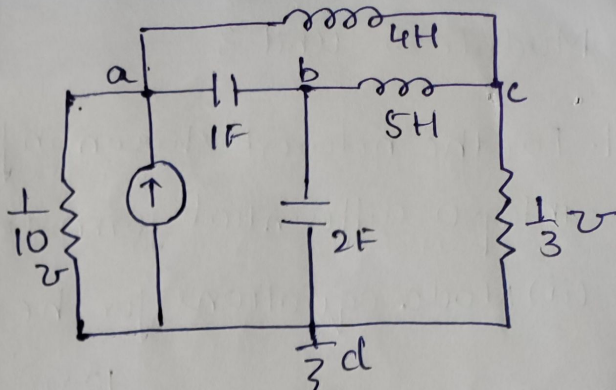
$$0 = \frac{1d}{dt}(i_2 - i_1) + \frac{2d}{dt}(i_2) + \frac{1}{5}\int (i_2 - i_3)\,dt.$$

KVL for mesh 3,

$$0 = \frac{1}{4}\int (i_3 - i_1)\,dt + \frac{1}{5}\int (i_3 - i_2)\,dt + 3i_3.$$

Dual of the network given:





Dual network.

(ii) Node Equations:

Applying KCL for node ⓐ,

$$10 = (V_a)10 + \frac{1}{4}\int (V_a - V_c)\,dt + \frac{1\,d(V_a - V_b)}{dt}.$$

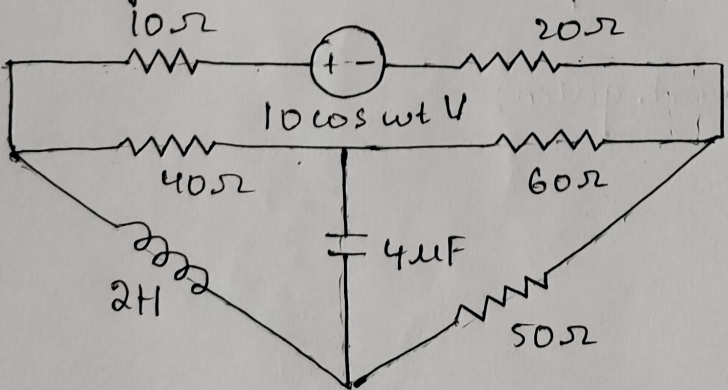
Applying KCL for node ⓑ,

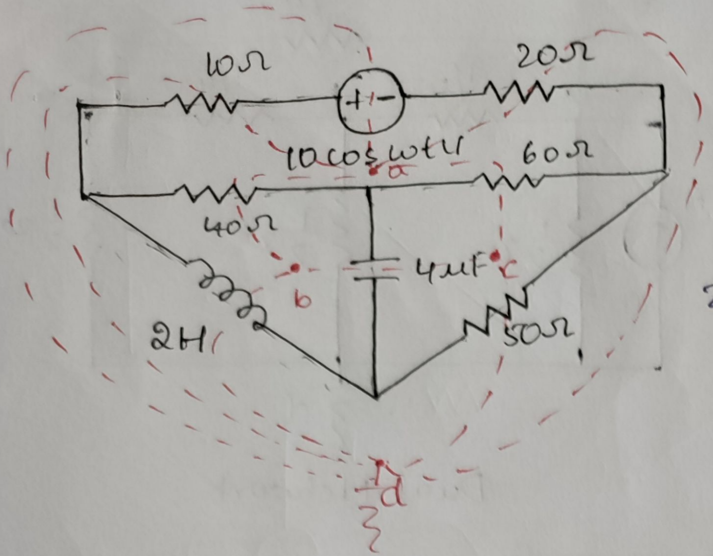
$$0 = \frac{1\,d(V_b - V_a)}{dt} + \frac{2\,dV_b}{dt} + \frac{1}{5}\int (V_b - V_c)\,dt.$$

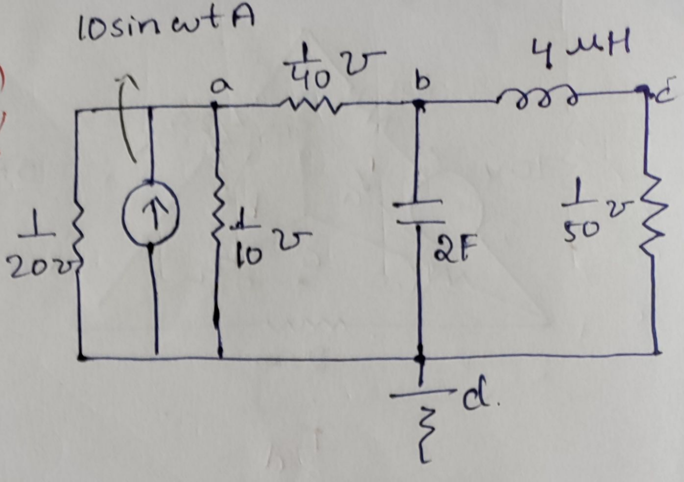
Applying KCL for node ⓒ,

$$0 = (V_c)3 + \frac{1}{4}\int (V_c - V_a)\,dt + \frac{1}{5}\int (V_c - V_b)\,dt.$$

2. Draw the dual of the network shown in figure

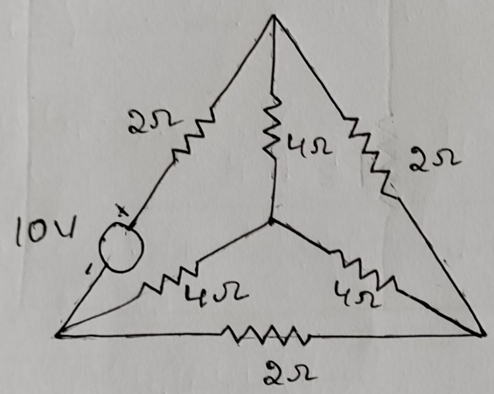






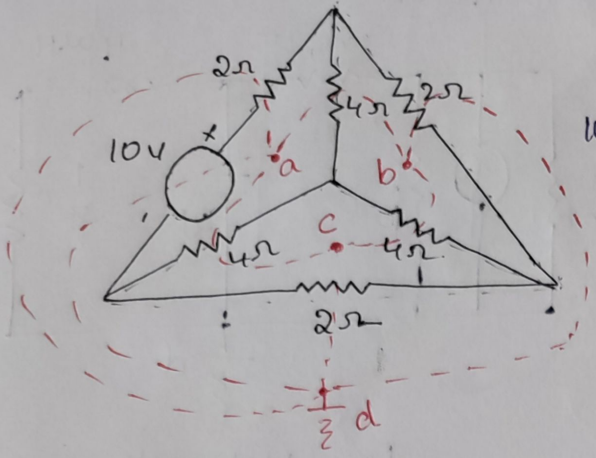
Dual Network.

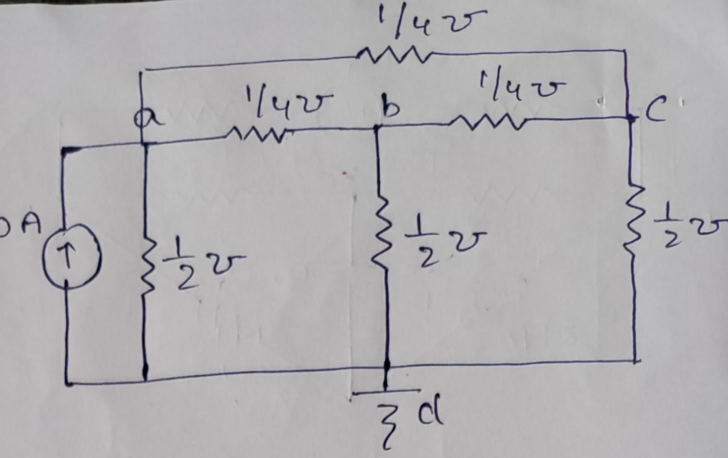
3. What are dual Networks ? Write their significance ? Draw the dual of the circuit shown in Fig.



Two networks are said to be dual if the mesh Equations of one network corresponds to the node equations of other Network. The KCL statement is similar to KVL statement when current is replaced by voltage and node is replaced by mesh.

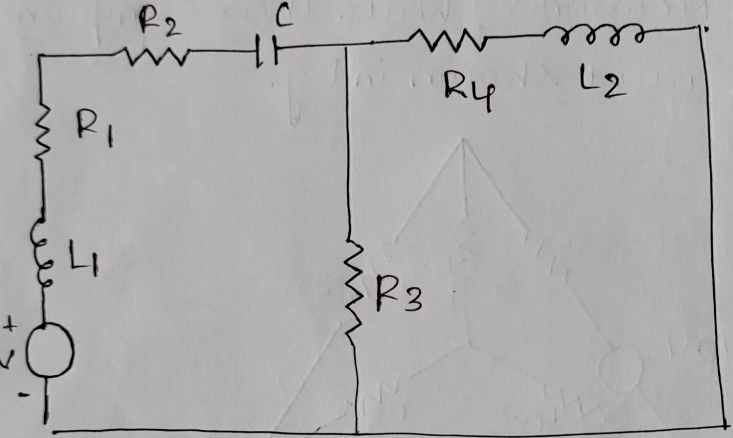
Significance : Duality helps in simplifying analysis and design of electrical networks by transforming one network into another mathematically equivalent dual network.

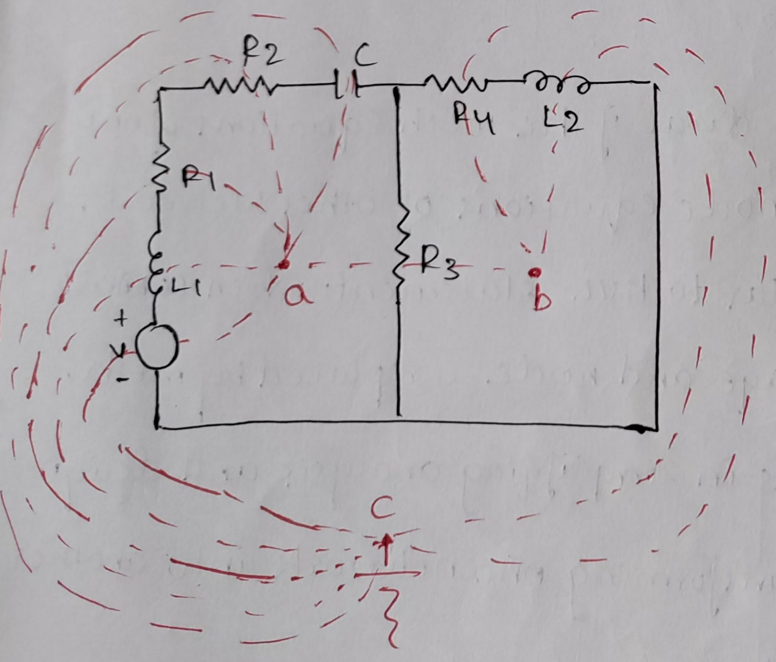


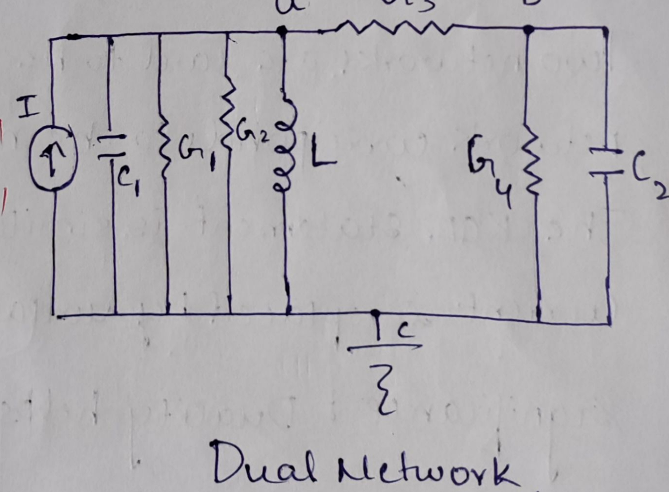


Dual Network.

4. Draw the dual for the circuit shown in figure.

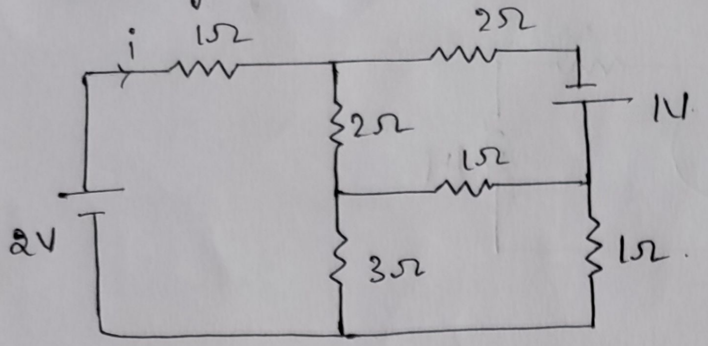


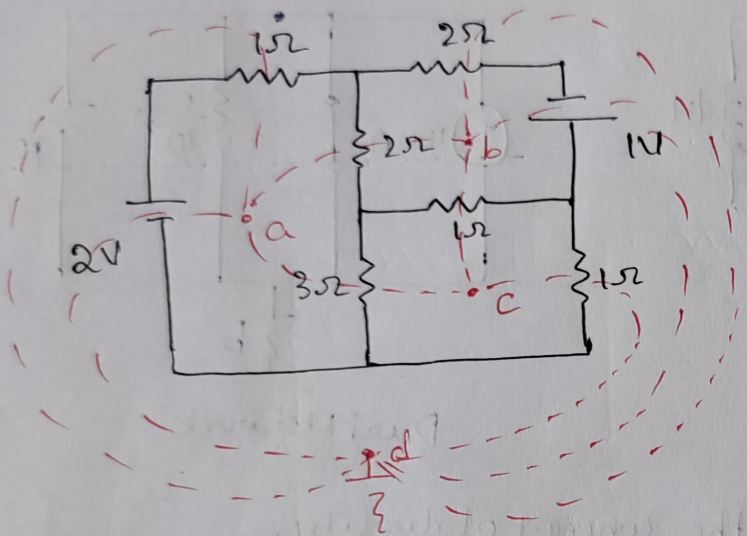


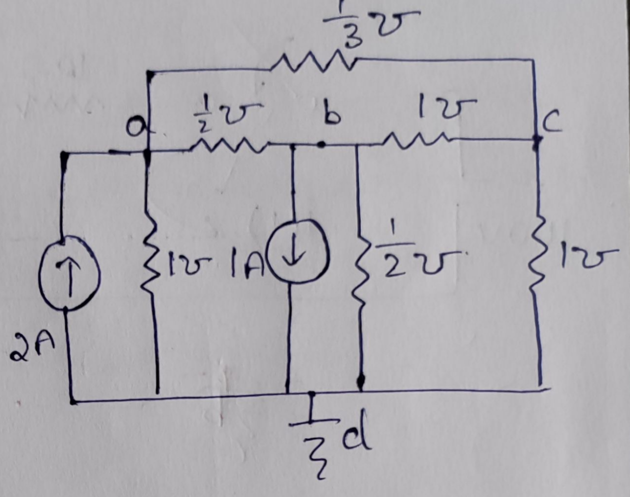


Dual Network.

5. Draw the dual for the circuit shown in figure :

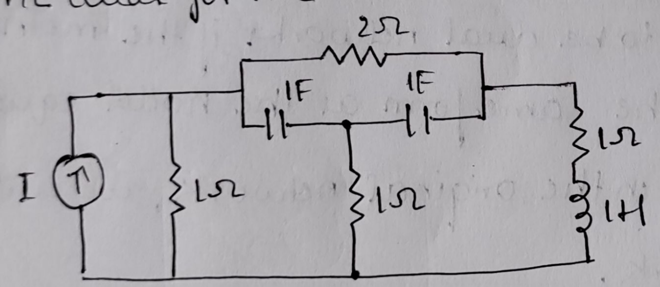


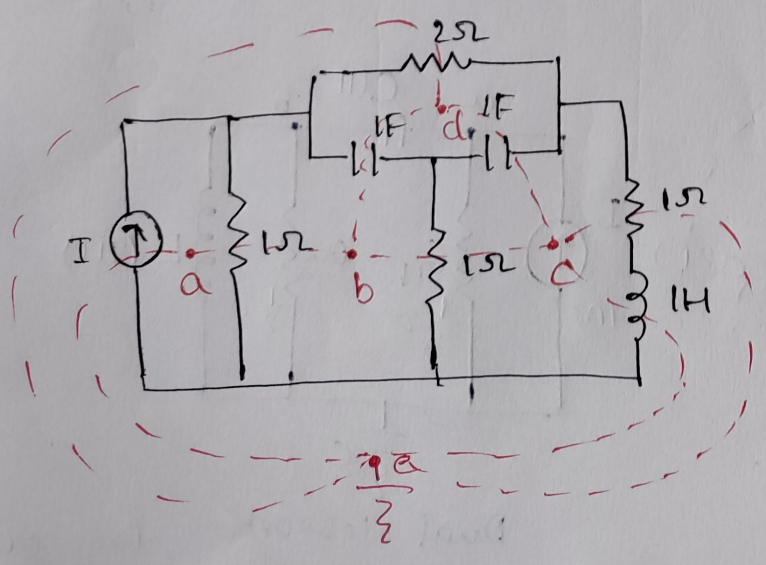


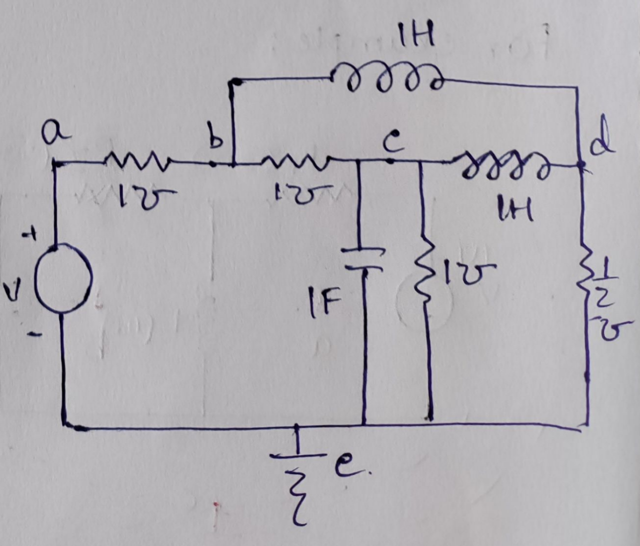


Dual Network.

6. Draw the dual for the circuit shown in figure .

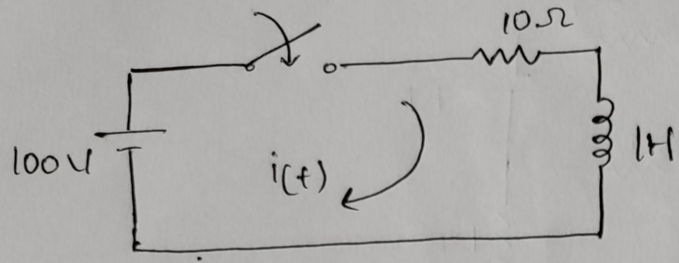


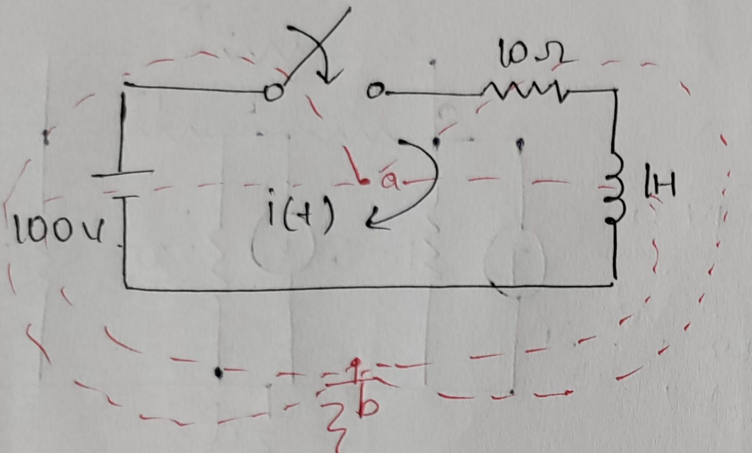


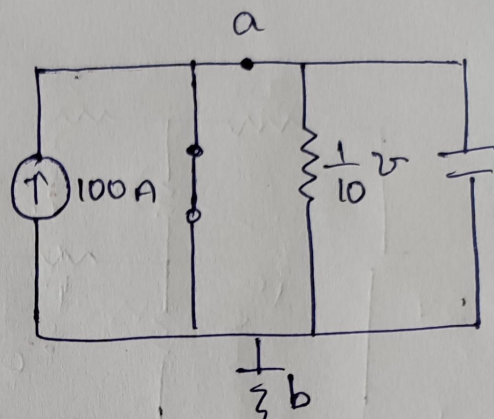


Dual Network.

7. Draw the dual for the circuit shown in figure.





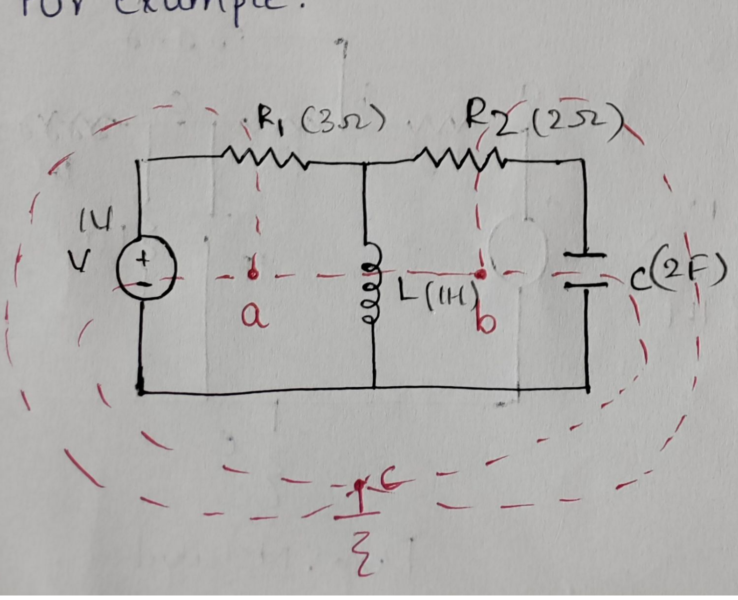


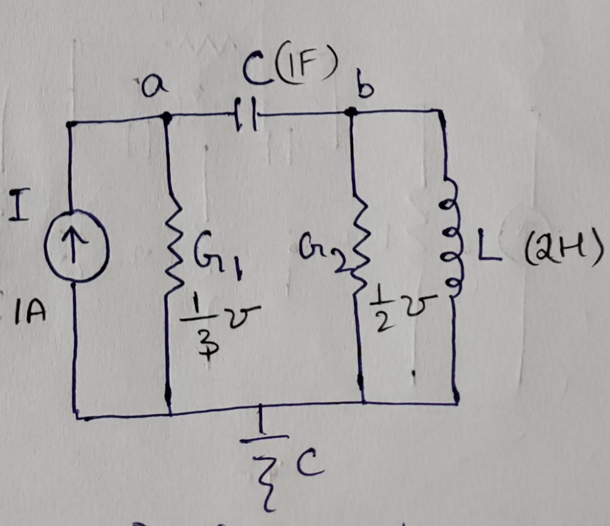
Dual Network.

8. With an example, explain the concept of duality.

Two networks are said to be dual networks if the mesh equations of one network have the same form as the nodal equations of the other. For every KVL in the original network, we will have a KCL in the dual network.

For example:





Dual Network

Here in the above example we can say that the voltage, Resistances, inductance and capacitance in the original network appear in dual network as Current, Conductances, Capacitance and inductance respectively with the same numerical values. The Dual pairs are voltage → Current, Resistance (R) → Conductance (G), Mesh → Node, Inductor → Capacitor, Capacitor → Inductor, Switch open → Switch close, and Series Path → Parallel Path. The Dual networks contains two different equations mesh and Nodal equations. The equations are termed as Integro differential Equations. The Mesh Equation / KVL Equation is,

$$v(t) = v_R + v_L + v_C$$

$$v(t) = Ri(t) + L\frac{di(t)}{dt} + \frac{1}{C}\int i(t)dt.$$

and The Node Equation / KCL Equation is,

$$i(t) = i_1 + i_2 + i_3$$

$$i(t) = v_1 G + C\frac{dv_1}{dt} + \frac{1}{L}\int v_1 dt$$

Dual	Pairs
Resistance (R)	Conductance (G)
Inductance (L)	Capacitance (C)
Voltage (V)	Current (I)
Voltage source	current source
Node	Mesh
Series path	parallel path
Open circuit	Short circuit
KVL	KCL